A finite-sum Lyapunov function for a stocked Ricker map

Abstract

We consider the fixed planar map $T(x,y) = (x\exp(3/2-y)+\mathrm{e}-1, x)$, an example in the literature on delayed Ricker models with stocking. Every positive original phase enters an explicit invariant rectangle after thirty iterates. A positive quadratic function contracts there after five iterates, as proved by a finite exact-integer interval certificate and an analytic local estimate. A weighted sum of thirty-five evaluations of that quadratic function is then a real analytic, coercive state Lyapunov function with strict decrease on the entire positive quadrant. This proves global asymptotic stability, square summability of the phase error and an explicit local-entry bound. We also prove that no centered constant positive-definite quadratic can be nonincreasing at every step on the source-associated invariant square. The finite-step conversion and rigorous verification tools are classical; the result is a quantitative verification for this one prescribed map.

Keywords: Lyapunov function; discrete dynamical system; delayed Ricker map; global stability; rigorous computation

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1 Introduction

For a delayed recurrence, the Lyapunov function must control its full phase. Local stability and an invariant rectangle can leave a substantial gap: an energy that contracts close to the equilibrium may increase during a nonlocal excursion. A decrease over several actual iterates can bridge this gap, provided the intermediate states and the whole verification region are controlled.

We study

$$T(x,y) = \big(x\exp(3/2-y)+c, x\big), \qquad c = \mathrm{e}-1, \qquad D = (0,\infty)^2. \tag{1}$$

The original coordinates are arbitrary and independent. With $s_0 = (x_0, x_{-1}) \in D$, write

$$s_n = T^n(s_0) = (x_n, x_{n-1}), \qquad x_{n+1} = x_n\exp(3/2 - x_{n-1}) + c. \tag{2}$$

This is equation (1.3) of AlSharawi–Kallel [1], at the stable parameter choice $r = 3/2$, $h = \mathrm{e}-1$ in their Example 3.3. Here h denotes stocking in that source, not a numerical step size. The map is autonomous; no differential equation or time-discretization limit is used.

For the general stocked map in [1], define

$$H^* = \frac{h+\sqrt{h^2+4h}}{2}, \qquad r_2 = H^* + \log(H^*-h) - \log H^*, \qquad r_1 = h+1-\log(h+1).$$

Their Theorem 3.6 gives a sufficient global-attraction range and, in a wider range, bounds described by two pseudo-fixed points of an embedding. At the parameters in (1), $r_2 < 3/2 < r_1 = h$. The broader local-to-global question is discussed in the conclusion of the identified author version. We use its concrete example, without making a claim about the present status of that broader question.

The later same-map example in AlSharawi–Cánovas [2, Example 4.12] retains the sufficient condition $q < H^*$. The original-map hypothesis of their Theorem 4.11 is stronger than local eigenvalue stability and does not hold at (1). The stocking application in [3, Section 5.1] has delay two and a different phase dimension. Exact correspondence and access limits are recorded in the companion verification note.

Finite-step Lyapunov functions and their conversion to one-step functions by finite orbit sums have established antecedents [4]. Bobiti–Lazar [5] combine such sums with rigorous cellwise verification and a local Lyapunov core. Computer-assisted stability proofs also have precedents for unstocked delayed Ricker dynamics [6]. The purpose here is to give a reproducible certificate for (1), together with the analytic capture and stability arguments needed to turn it into an all-positive-phase result.

Put $p = (q, q)$, where q is the unique positive solution of

$$q = q\exp(3/2 - q) + c. \tag{3}$$

Define

$$\begin{gathered} L_0 = \frac{859}{500}, \qquad U_0 = \frac{44}{5}, \qquad B = [L_0, U_0]^2, \\ P = \begin{pmatrix} 8/7 & -5/28 \\ -5/28 & 1 \end{pmatrix}, \qquad V(s) = (s-p)^T P (s-p). \end{gathered} \tag{4}$$

The two finite state functions used below are

$$W(s) = \sum_{j=0}^{4} V(T^j s), \qquad Z(s) = \sum_{j=0}^{29} V(T^j s) + 10 \sum_{j=30}^{34} V(T^j s). \tag{5}$$

These definitions involve a fixed number of evaluations of the map from the current phase. They contain neither a variable entrance time nor an infinite tail.

Theorem 1. *For the map* (1)*, the following statements hold.*

(i) *The positive quadrant is invariant, and p is its unique fixed point. Moreover,*

$$\frac{8591}{5000} < c < \frac{43}{25}, \qquad \frac{12947}{5000} < q < \frac{5179}{2000}. \tag{6}$$

The rectangle B is forward invariant, and

$$T^{30}(s_0) \in B \qquad (s_0 \in D). \tag{7}$$

(ii) *The quadratic form satisfies*

$$\frac{23}{28} I \preceq P \preceq \frac{37}{28} I. \tag{8}$$

On B it obeys the five-step inequality

$$V(T^5 s) \le \frac{9}{10} V(s), \qquad V(s) < 102. \tag{9}$$

The set

$$E = \{s \in \mathbb{R}^2 : V(s) \le 23/70000\} \tag{10}$$

is invariant. On E, and more generally whenever $|x-q|, |y-q| \le 1/50$, one has

$$V(Ts) \le V(s) - \frac{1}{20}\|s-p\|^2 \le \frac{178}{185} V(s). \tag{11}$$

(iii) *The functions V, W, Z are real analytic and nonnegative on $\mathbb{R}^2$, with zero set exactly $\{p\}$. Both W and Z are coercive and dominate V. Their stated descent domains are*

$$W(Ts) - W(s) \le -\frac{1}{10} V(s) \quad (s \in B), \qquad Z(Ts) - Z(s) \le -V(s) \quad (s \in D). \tag{12}$$

Thus p is globally asymptotically stable relative to D. For every $s_0 \in D$,

$$\sum_{n=0}^{\infty} V(s_n) \le Z(s_0), \qquad \sum_{n=0}^{\infty} \|s_n - p\|^2 \le \frac{28}{23} Z(s_0). \tag{13}$$

For $s \in B$ and every integer $m \ge 0$,

$$V(T^{5m} s) \le (9/10)^m V(s). \tag{14}$$

Every original positive orbit belongs to E by index 635 and remains there thereafter.

The constants are sufficient and deliberately conservative. Descent on negative phases, a sharp entrance bound and a parameter neighborhood are not asserted. The five-step inequality is a computer-assisted statement with a complete exact certificate; its analytic and computational components are given below.

2 Capture from the original positive phase

Lemma 1. *Statements* (i) *and* (7) *of Theorem 1 hold, and all iterates following index 30 remain in B.*

Proof. Positivity follows directly from $c > 0$. A fixed point has equal coordinates. Let

$$g(t) = t\big(1 - \exp(3/2 - t)\big) - c.$$

For $0 < t \le 3/2$ this is negative. For $t > 3/2$,

$$g'(t) = 1 + (t-1)\exp(3/2 - t) > 0,$$

and $g(t) \to \infty$. Hence there is exactly one positive root. The exact exponential bounds described in Section 4 verify (6) by the signs of g at the two displayed endpoints.

The original first two updates give $x_1 > c$, $x_2 > c$ and

$$\begin{aligned} x_2 &= x_0 \exp(3 - x_{-1} - x_0) + c \exp(3/2 - x_0) + c \\ &< \mathrm{e}^2 + c\mathrm{e}^{3/2} + c < 17. \end{aligned}$$

Here $t\mathrm{e}^{-t} \le \mathrm{e}^{-1}$ for $t > 0$ and $x_{-1} > 0$. The last strict inequality is an exact scalar check. In particular, this bound has no assumption on the sizes or relative sizes of the original two coordinates.

For every $n \ge 2$, $x_{n-1} > c$ and

$$\exp(3/2 - c) < \frac{201}{250}, \qquad x_{n+1} \le \frac{201}{250} x_n + \frac{43}{25}. \tag{15}$$

Iteration from $x_2 < 17$ gives

$$x_{2+m} \le \frac{430}{49} + \left(\frac{201}{250}\right)^m \left(17 - \frac{430}{49}\right), \qquad m \ge 0. \tag{16}$$

The affine limiting level satisfies $430/49 < 44/5$, and exact rational arithmetic gives

$$\frac{430}{49} + \left(\frac{201}{250}\right)^{27} \left(17 - \frac{430}{49}\right) < \frac{44}{5}.$$

Therefore $x_{29} < U_0$ and, by (15),

$$x_{30} < \frac{201}{250}\frac{44}{5} + \frac{43}{25} = \frac{5497}{625} < U_0.$$

Both coordinates exceed $c > L_0$. This proves precisely $T^{30} s_0 = (x_{30}, x_{29}) \in B$.

For $(x, y) \in B$, monotonicity in x and $-y$ gives

$$c < x \exp(3/2 - y) + c \le U_0 \exp(3/2 - L_0) + c < U_0.$$

The final strict inequality is verified by the same exact scalar bounds, with margin greater than $1/200$. The second output coordinate is $x \in [L_0, U_0]$. Thus B is invariant, including its boundary. This proof establishes capture and invariance before the five-step estimate is used. ∎

3 The local quadratic estimate

Lemma 2. *The matrix bounds* (8)*, the local estimate* (11) *and invariance of* E *hold. In particular,*

$$V(T^5 s) \le (178/185)^5 V(s) \le (9/10) V(s) \qquad (s \in E). \tag{17}$$

Proof. The bounds on P follow from $2|uv| \le u^2 + v^2$, or from its row sums. On the square $|x-q|, |y-q| \le 1/50$, write

$$DT(x,y) = \begin{pmatrix} \alpha & \beta \\ 1 & 0 \end{pmatrix}, \qquad \alpha = \exp(3/2 - y), \quad \beta = -x\exp(3/2-y).$$

Let Q be the rigorous interval for q obtained by the sign-preserving bisection in Section 4. Evaluating x, y on $Q + [-1/50, 1/50]$ gives interval bounds for α, β. For every pair in these bounds, the symmetric matrix

$$M = P - \begin{pmatrix} \alpha & 1 \\ \beta & 0 \end{pmatrix} P \begin{pmatrix} \alpha & \beta \\ 1 & 0 \end{pmatrix} - \frac{1}{20} I$$

has entries

$$\begin{aligned} M_{11} &= \frac{8}{7} - \left(\frac{8}{7}\alpha^2 - \frac{5}{14}\alpha + 1 \right) - \frac{1}{20}, \\ M_{22} &= 1 - \frac{8}{7}\beta^2 - \frac{1}{20}, \\ M_{12} &= -\frac{5}{28} - \left(\frac{8}{7}\alpha - \frac{5}{28} \right) \beta. \end{aligned}$$

The exact interval evaluation verifies

$$M_{11} - |M_{12}| > \frac{3}{50}, \qquad M_{22} - |M_{12}| > \frac{1}{50}. \tag{18}$$

Hence M is positive definite. The bounds are valid for every pair of entries in the interval enclosure, without requiring their dependence on x, y.

For s in the square, the segment from p to s remains in it. The average of DT along that segment has the same entry bounds and second row $(1, 0)$. Since $T(p) = p$, the mean-value identity gives $T(s) - p = \overline{J}(s-p)$ with this average matrix. Consequently

$$V(Ts) \le V(s) - \frac{1}{20}\|s-p\|^2 \le \left(1 - \frac{1}{20}\frac{28}{37} \right) V(s),$$

which is (11). The lower bound on P implies that $s \in E$ satisfies

$$\|s-p\|^2 \le \frac{28}{23}\frac{23}{70000} = \frac{1}{2500}.$$

Thus E lies in the closed local square. Its invariance follows from (11), allowing the estimate to be iterated. Finally $(178/185)^5 < 9/10$ is an exact rational comparison. At p both sides of (17) vanish. ∎

4 The exact five-step certificate

Lemma 3. *The inequalities* (9) *hold on the entire closed rectangle* B.

Proof. We describe a finite exact verification, supplied in executable form with this note. Its program `verify_exact.py`, certificate `partition.jsonl`, and arithmetic documentation can be replayed using Python's standard library. The proof decisions use integers only. The local argument of Lemma 2 supplies the part where the decrement vanishes at the equilibrium.

Let $S = 2^{96}$. An integer interval $[l, u]$ represents $[l/S, u/S]$. A rational a/b, with $b > 0$, is enclosed by $[\lfloor Sa/b \rfloor, \lceil Sa/b \rceil]/S$. Addition is exact at the endpoints. Multiplication takes the minimum and maximum of the four endpoint products and rounds division by S outwards. Squaring assigns lower endpoint zero when the interval contains zero. For a zero-excluding interval, reciprocation uses

$$\left[l/S, u/S\right]^{-1} \subset \left[\left\lfloor \frac{S^2}{u} \right\rfloor, \left\lceil \frac{S^2}{l} \right\rceil\right] /S.$$

These rules include signed intervals; every required reciprocal excludes zero.

For a nonnegative dyadic endpoint t, choose an integer $k \geq 0$ so that $t/2^k \leq 1/16$, and round the reduced argument outwards. Its upper bound is at most $1/8$. Evaluate the positive exponential Taylor sum through degree 24 with outward rounding. For $0 \leq z \leq 1/8$ the omitted tail obeys

$$\sum_{j=25}^{\infty} \frac{z^j}{j!} \leq \frac{2(1/8)^{25}}{25!} < 2^{-96}. \tag{19}$$

The first bound follows because consecutive terms have ratio at most $1/(8 \cdot 26) < 1/2$. One dyadic unit added to the upper Taylor endpoint therefore encloses the tail. After k outward squarings the result encloses $\exp(t)$. Negative arguments use reciprocation, and monotonicity supplies interval endpoint bounds. The strict rational comparison in (19) is also checked by the program.

The interval for c is obtained from $\exp(1) - 1$. Starting with the rational bracket in (6), the interval for q is refined by dyadic bisection of g. An endpoint is changed only when the complete residual enclosure has a strict sign; an inconclusive test stops refinement without discarding the current bracket. The root is identified by the uniqueness argument in Lemma 1. The same arithmetic verifies the scalar capture inequalities and (18).

Set

$$F(s) = V(T^5 s) - \frac{9}{10} V(s).$$

The certificate begins with the dyadic outer rectangle

$$B_d = [\lfloor SL_0 \rfloor/S, \lceil SU_0 \rceil/S]^2 \supset B.$$

An unaccepted cell is bisected in its longer coordinate, choosing the first coordinate in a tie, at the exact integer midpoint. The two closed children cover their parent and share its midpoint face. The saved certificate specifies a finite binary path for every accepted

leaf. Replay reconstructs all cells from these paths and maintains the complete depth-first stack; a gap, duplicate, reordering, extra leaf or nonempty final stack is rejected.

There are three acceptance rules for a leaf $R = X \times Y$. First, the completed-square identity

$$V(x,y) = \frac{8}{7}\left(x - q - \frac{5}{32}(y-q)\right)^2 + \frac{871}{896}(y-q)^2 \tag{20}$$

gives an interval upper bound. If that bound is at most a downward-rounded 23/70000, then $R \subset E$ and Lemma 2 proves the claim. Second, direct interval evaluation of the five actual updates can give an upper bound for F that is nonpositive.

The third rule uses the mean-value estimate on a cell. Along an update, put $r = \exp(3/2 - y)$ and $z = xr$. If d_x, d_y are the derivative rows of the current two coordinates with respect to the input, then

$$(x,y) \longmapsto (z + c, x), \qquad (d_x, d_y) \longmapsto (r d_x - z d_y, d_x). \tag{21}$$

Start with $d_x = (1,0)$ and $d_y = (0,1)$ and apply (21) five times using interval arithmetic. For $u = x - q$, $v = y - q$, differentiation of the terminal quadratic gives

$$dV = 2\left[\left(\frac{8}{7}u - \frac{5}{28}v\right) d_x + \left(-\frac{5}{28}u + v\right) d_y\right].$$

Subtracting 9/10 of the initial gradient encloses both components of ∇F on the whole cell. If s_c is its integer-midpoint center, and h_x, h_y its maximal coordinate distances to the faces, the mean-value formula yields

$$F(s) \le \overline{F(s_c)} + \overline{|\partial_x F|}\, h_x + \overline{|\partial_y F|}\, h_y \qquad (s \in R). \tag{22}$$

Bars denote rigorous interval upper bounds, with upward rounding in the products. A nonpositive right side certifies the cell. This computes derivatives of the actual fivefold composition, rather than derivatives along a numerical trajectory.

The supplied exhaustive replay certifies 641 closed leaves: 18 by the local-sublevel rule, 47 by direct intervals and 576 by (22). The maximum binary depth is 22, and the final covering stack is empty. Every recorded acceptance margin is strictly positive. A local-cell margin means strict inclusion in E; it does not claim a strictly negative value of $F(p)$. All root-box faces and shared subdivision faces are included. Thus every point of B satisfies $F \le 0$. The computer-assisted part is precisely this finite set of rational integer decisions with the proved exponential bounds.

Finally, V is convex. On the rectangle it is at most the largest value at its four vertices. The same exact scalar evaluation verifies $102 - V > 33$ at each vertex, proving $V < 102$ throughout B. ∎

5 The global finite-sum function and actual orbits

Proof of Theorem 1. Lemmas 1–3 prove parts (i) and (ii). Since T is real analytic on $\mathbb{R}^2$, each finite composition and each function in (5) is real analytic there. Positive definiteness of P gives $V \geq 0$ with zero set $\{p\}$. The sums have positive coefficients, include V and vanish at p, so they have the same zero set and satisfy

$$W(s), Z(s) \geq V(s) \geq \frac{23}{28}\|s-p\|^2 \qquad (s \in \mathbb{R}^2).$$

In particular, they are coercive.

Write $V_j = V(T^j s)$. The exact telescoping identities are

$$W(Ts) - W(s) = V_5 - V_0, \qquad Z(Ts) - Z(s) = -V_0 - 9V_{30} + 10V_{35}. \tag{23}$$

On B, the first identity and (9) give the asserted loss for W. If $s \in D$, Lemma 1 gives $T^{30}s \in B$, so the same five-step estimate gives $V_{35} \leq (9/10)V_{30}$. The second identity proves the strict loss for Z on D. Capture is used here only after it has been proved independently.

Since D is invariant, summation along the original orbit gives

$$\sum_{n=0}^{N-1} V(s_n) \leq Z(s_0) - Z(s_N) \leq Z(s_0).$$

Taking increasing limits and using (8) proves (13). In particular $V(s_n) \to 0$ and $s_n \to p$.

For completeness, fix $\varepsilon > 0$. Continuity of Z at p, with $Z(p) = 0$, gives $\delta > 0$ such that $\|s-p\| < \delta$ implies $Z(s) < (23/28)\varepsilon^2$. For $s \in D$ satisfying this condition, every iterate obeys

$$\frac{23}{28}\|T^n s - p\|^2 \leq Z(T^n s) \leq Z(s) < \frac{23}{28}\varepsilon^2.$$

Thus p is Lyapunov stable relative to D. The attraction proved above is global on this same domain.

Invariance of B permits iteration of (9), proving (14). For arbitrary $s_0 \in D$, one has $V(T^{30}s_0) < 102$. The exact rational inequality

$$102\left(\frac{9}{10}\right)^{121} < \frac{23}{70000}$$

therefore implies $T^{635}s_0 \in E$, because $635 = 30 + 5 \cdot 121$. Invariance of E completes the proof. ∎

6 A limited obstruction to one-step quadratics

The pseudo-fixed points used in [1] describe a useful invariant square. They are associated with an embedding and are not asserted to be an actual periodic orbit of T. We identify the required pair independently.

Proposition 6.1. Let $f(t) = c/(1 - \exp(3/2 - t))$ for $t > 3/2$. There is a unique

$$a \in \left(\frac{3439}{2000}, \frac{4299}{2500}\right) \quad \text{such that} \quad f(f(a)) = a.$$

Put $b = f(a)$ and $K = [a, b]^2$. Then $a < q < b$, $K \subset B$ and K is invariant. No constant real symmetric positive-definite matrix H satisfies

$$(Ts - p)^T H(Ts - p) \le (s - p)^T H(s - p) \qquad (s \in K). \tag{24}$$

Proof. The function f is decreasing and takes values greater than $c > 3/2$. On

$$I = [3439/2000, 4299/2500], \qquad J = [871/100, 873/100],$$

exact exponential endpoint enclosures give $f(I) \subset \operatorname{int} J$ and $f(J) \subset \operatorname{int} I$. More explicitly,

$$\begin{aligned} 8.7187 < f(3439/2000) < 8.7188, \qquad & 8.7151 < f(4299/2500) < 8.7152, \\ 1.71955 < f(871/100) < 1.71956, \qquad & 1.71952 < f(873/100) < 1.71953. \end{aligned}$$

Every terminating decimal here denotes an exact rational endpoint. These four bounds, or the tighter composition bounds supplied by the public checker, follow from the Taylor enclosure of Section 4.

The same endpoint bounds give $\exp(3/2 - 3439/2000) < 81/100$ and $\exp(3/2 - 871/100) < 1/1000$. Since

$$|f'(t)| = \frac{c \exp(3/2 - t)}{(1 - \exp(3/2 - t))^2},$$

one has $|f'| < 40$ on I and $|f'| < 1/500$ on J. Therefore $f \circ f$ maps I into its interior and has derivative strictly between 0 and $2/25$. The endpoint signs and the intermediate value theorem give exactly one root in I. This is uniqueness in the stated interval; q is another fixed point of $f \circ f$ outside it.

Set $b = f(a)$. Then $8.71 < b < 8.73$, $f(b) = a$ and

$$a = a \exp(3/2 - b) + c, \qquad b = b \exp(3/2 - a) + c.$$

The root bounds imply $a < q < b$ and $K \subset B$. On K, the first coordinate of T attains its minimum at (a, b) and maximum at (b, a); the two identities give values a and b. Its second coordinate is in $[a, b]$. Thus K is invariant.

Suppose $H = \left(\begin{smallmatrix} h & k \\ k & \ell \end{smallmatrix}\right)$ is positive definite and satisfies (24). Put

$$\alpha = q - a > 0, \qquad r_0 = \exp(3/2 - q), \qquad d = q - c = qr_0, \qquad z = \sqrt{\ell/h}.$$

Positive definiteness gives $h > 0$, $\ell > 0$ and $k/h > -z$. At the phase (a, q), the initial displacement is $(-\alpha, 0)$ and the image displacement is $(-r_0\alpha, -\alpha)$. Therefore

$$r_0^2 + 2r_0 k/h + z^2 \le 1, \qquad (r_0 - z)^2 < 1, \qquad z < 1 + r_0.$$

At (q, a), the displacements are $(0, -\alpha)$ and $(d(\mathrm{e}^{\alpha} - 1), 0)$, respectively. Hence

$$z \geq d\frac{\mathrm{e}^{\alpha} - 1}{\alpha}.$$

These requirements are incompatible. The weaker bounds $129/50 < q < 13/5$, already implied by (6), give $\alpha > 43/50$ and $d > 43/50$. With $t = 43/50$, positive Taylor terms imply

$$d\frac{\mathrm{e}^{\alpha} - 1}{\alpha} > t\left(1 + \frac{t}{2} + \frac{t^2}{6} + \frac{t^3}{24}\right) = \frac{67930067}{50000000} > \frac{67}{50}.$$

On the other hand, $r_0 = 1 - c/q$ and $c > 859/500$ yield

$$1 + r_0 = 2 - c/q < 2 - \frac{859/500}{13/5} = \frac{1741}{1300} < \frac{67}{50}.$$

The contradiction proves the proposition. ∎

Proposition 6.1 concerns only centered constant positive quadratics under one-step nonincrease on K. The finite-step quadratic estimate and the analytic functions W, Z provide positive results on the same actual dynamics. In particular, the proposition is not an obstruction to general Lyapunov functions or to trajectory convergence.

7 Verification and source scope

The public computation is a finite certificate of a universal inequality. Its arithmetic rules, exponential remainder, complete covering and analytic local core are part of the proof; a grid of sampled trajectories would not suffice. The companion includes the entire partition, a deterministic replay, exact scalar checks and supplementary arithmetic tests. The analytic arguments above establish positivity, original-index capture, the local estimate, the finite-sum cancellation and all infinite-orbit conclusions.

The orbit-sum construction (5) uses standard finite-step machinery. The particular weights in Z cancel the certified post-capture block, as shown in (23). Neither this cancellation nor the interval/local-core strategy is presented as a new general principle. The content is the explicit verified instance for the prescribed stocked map.

The comparison with [1–3] uses the stated arXiv versions. The final journal bodies of [1, 2] were not inspected, and no identity of their complete final contents with the author versions is presumed. For [4], the finite-sum attribution is corroborated by the directly inspected primary equations (6)–(7) of [5]; final-journal theorem numbering is not used here. The bounded literature comparison leaves priority qualified. No firstness claim or general local-to-global Ricker conjecture resolution is made.

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